



MECH WELL
DRIVEN BY INNOVATION



Date: 5th Aug 2026

Report No.: MWI/FEA/ACK-6148/01

FEA Report

Static Structural Analysis of Blade

Client:



1. Material Properties

Material	Item	Young's Modulus (Mpa)	Poisson's Ratio	Density (t/mm ³)	Yield Stress (Mpa)	Allowable Von-mises Stress (Mpa)	Allowable Shear Stress (Mpa)
IS 2062	Blade	175000	0.29	7.8E-9	183	140	-
SS 410	Shaft	175500	0.3	7.8E-9	223	171	85
EN 8	Dowel pin	190000	0.3	7.8E-9	240	184	92

Units : Length= mm, Mass= Tones, Time= Second, Force= Newton

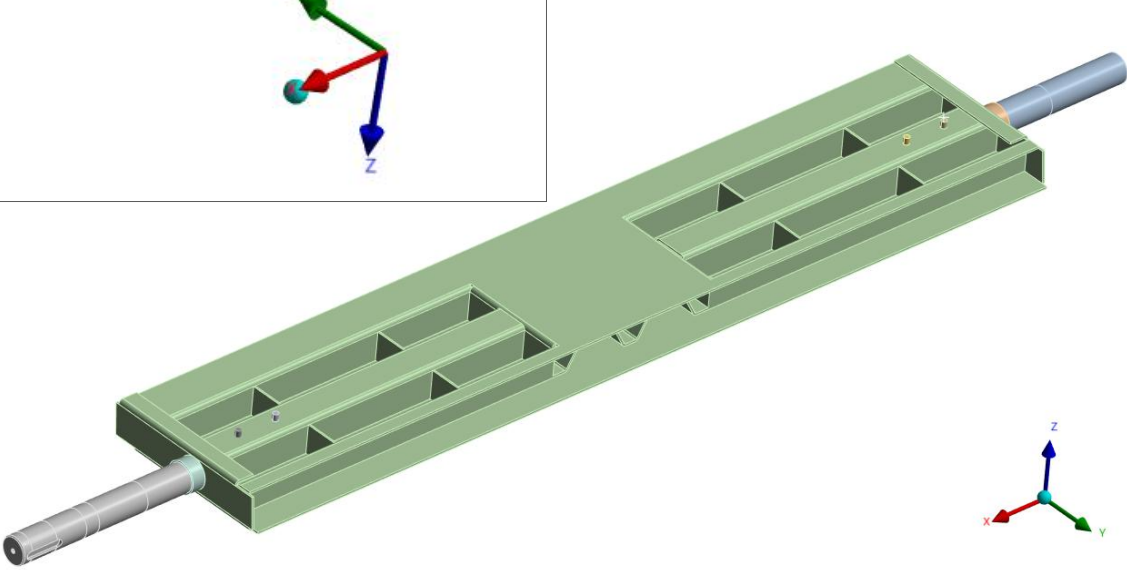
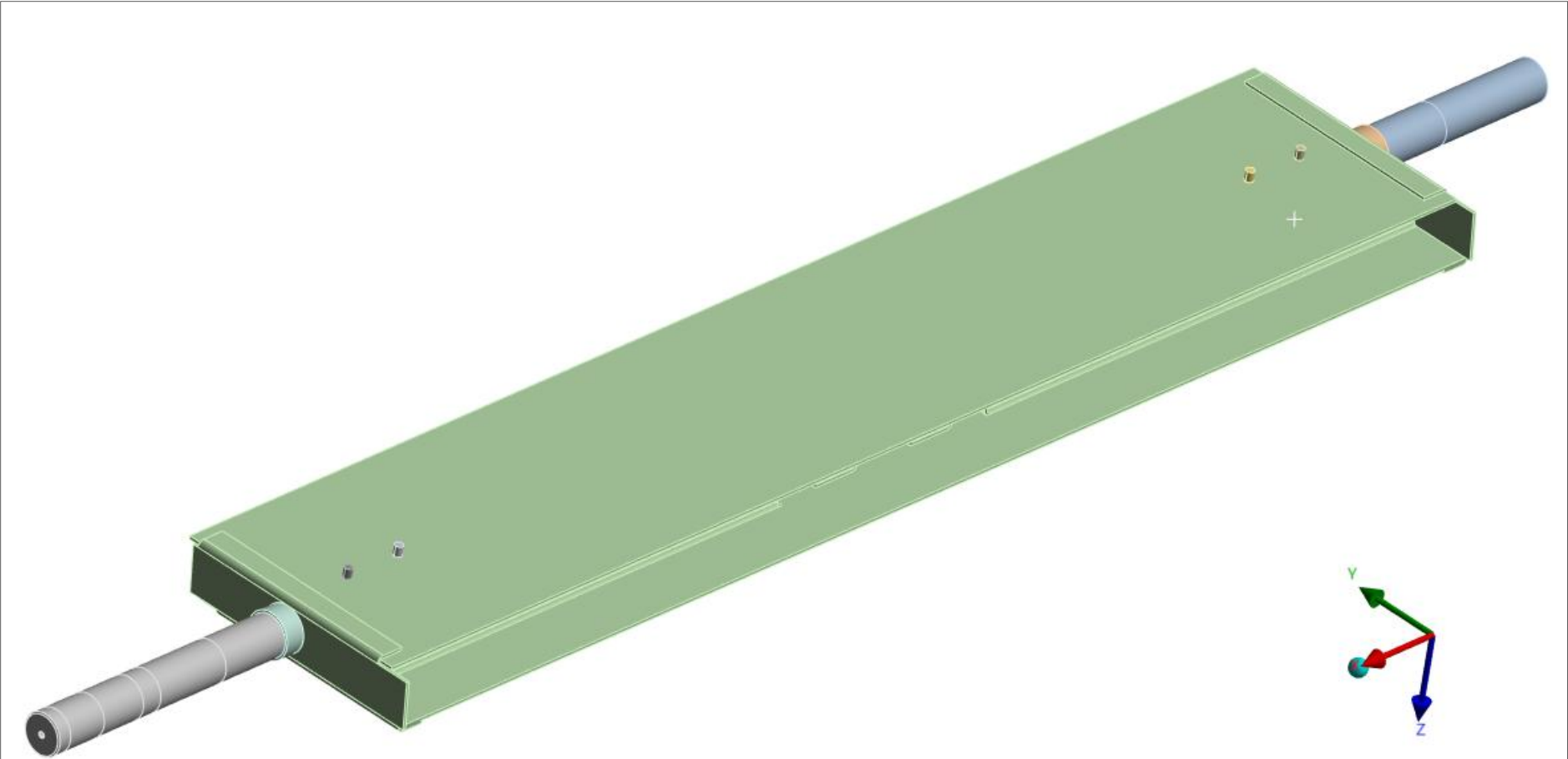
Allowable Von-mises Stress for Blade :-

$$\sigma_{all} = S_{yt} / FOS, \quad FOS = 1.3,$$

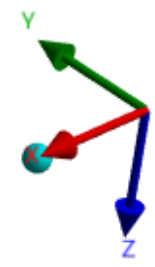
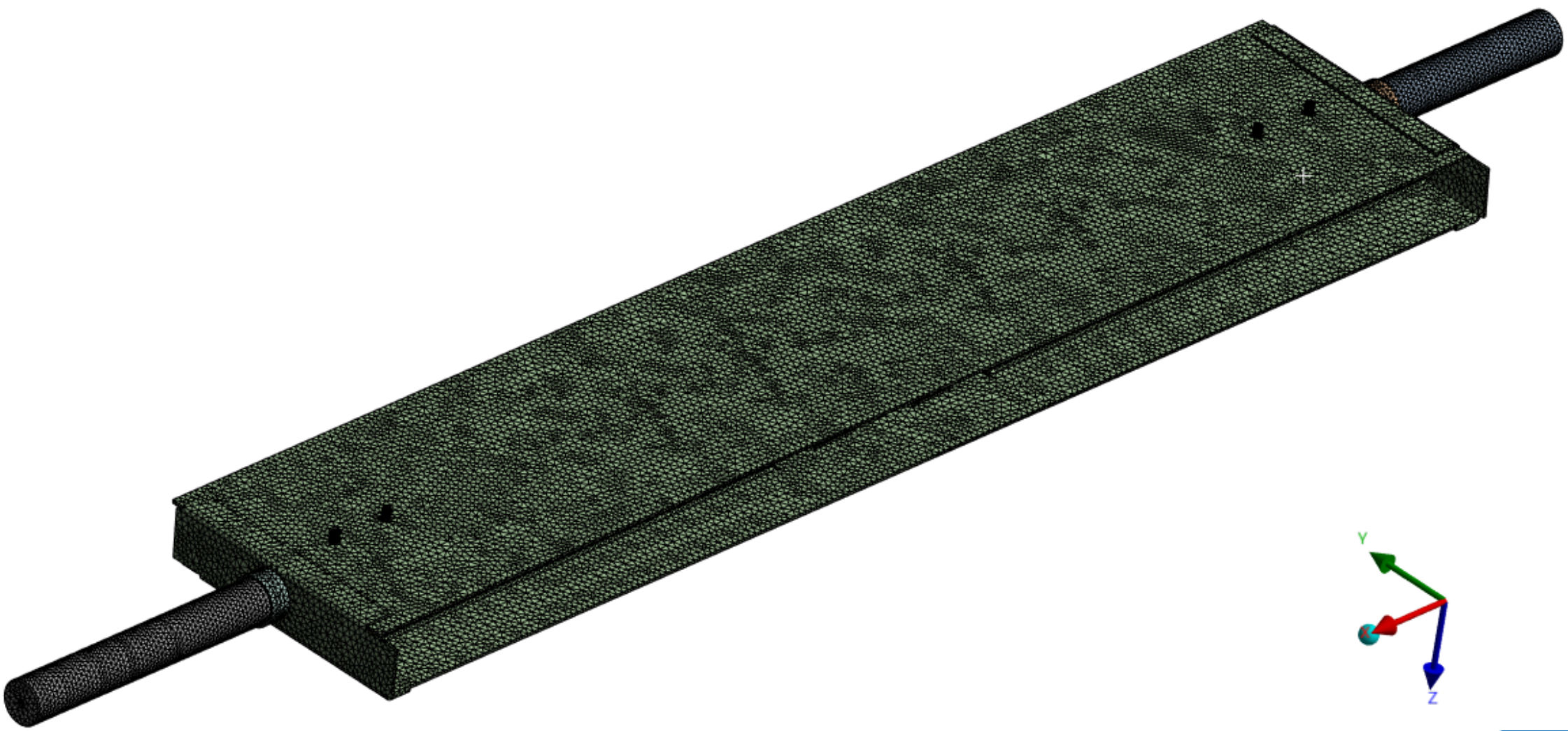
Allowable Max. Shear Stress for Shaft :-

$$\tau_{max} = 0.5 \times S_{yt} / 1.3$$

2. Geometry :



3. Mesh :





Damper Blade

Case-1 : Self Weight + pressure



4. Boundary Conditions :

B: 6148 blade self weight+pressure

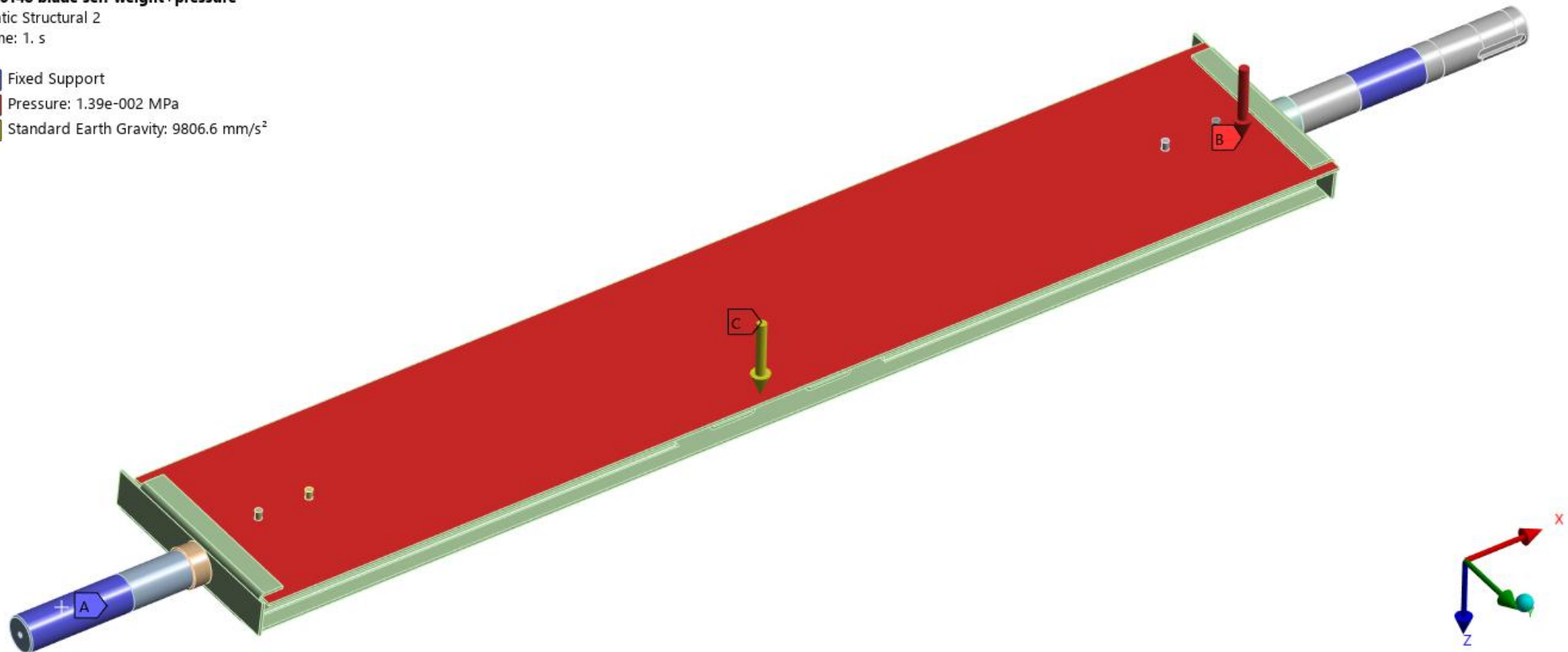
Static Structural 2

Time: 1. s

A Fixed Support

B Pressure: 1.39e-002 MPa

C Standard Earth Gravity: 9806.6 mm/s²

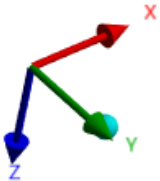
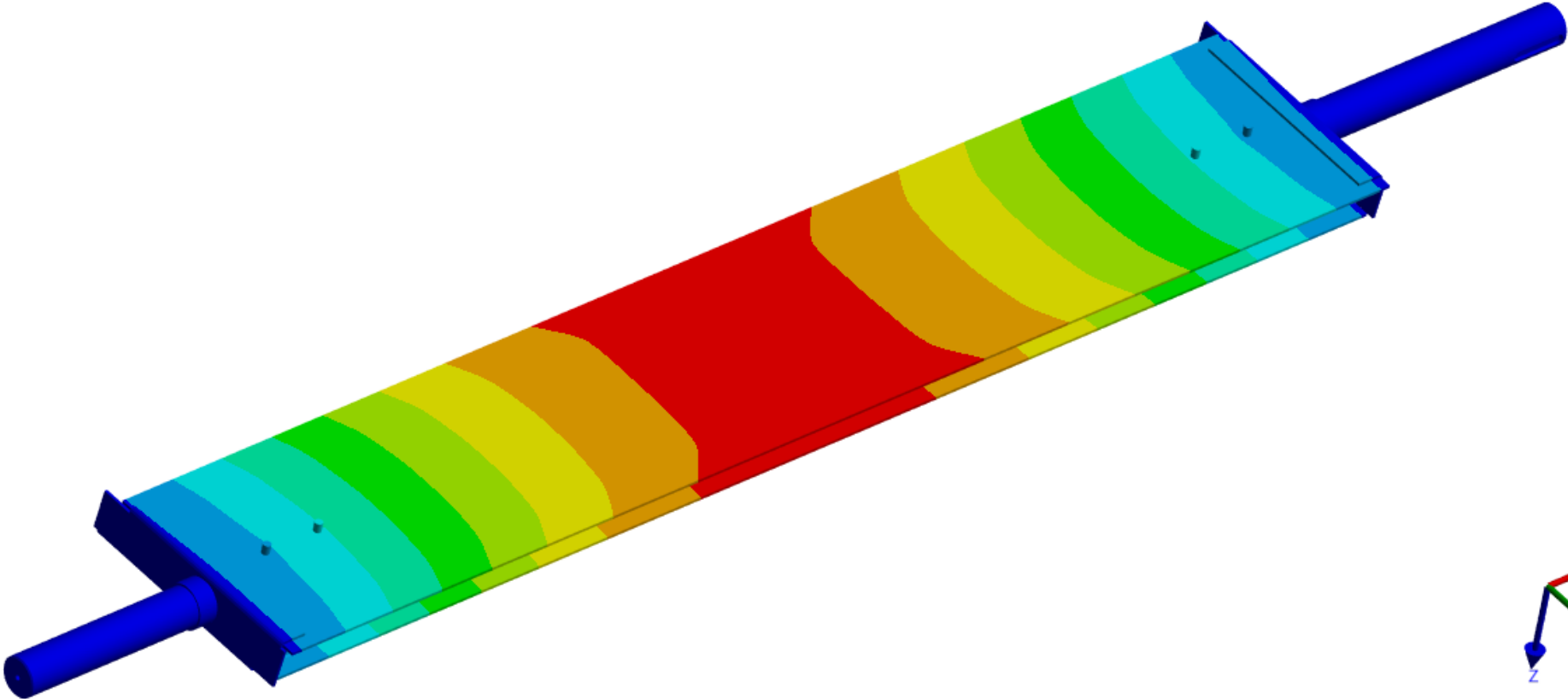
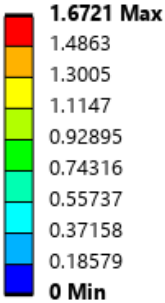


- A) Both shafts are fixed.
- B) Design Pressure 1421.92 mmWc which is 0.0139Mpa.
- C) Self weight of Blade.

5. Results : Deformation Plot



B: 6148 blade self weight+pressure
Total Deformation
Type: Total Deformation
Unit: mm
Time: 1 s



Max. Total Deformation = 1.7 mm



5.1 Results : Von-mises Stress Plot

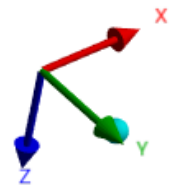
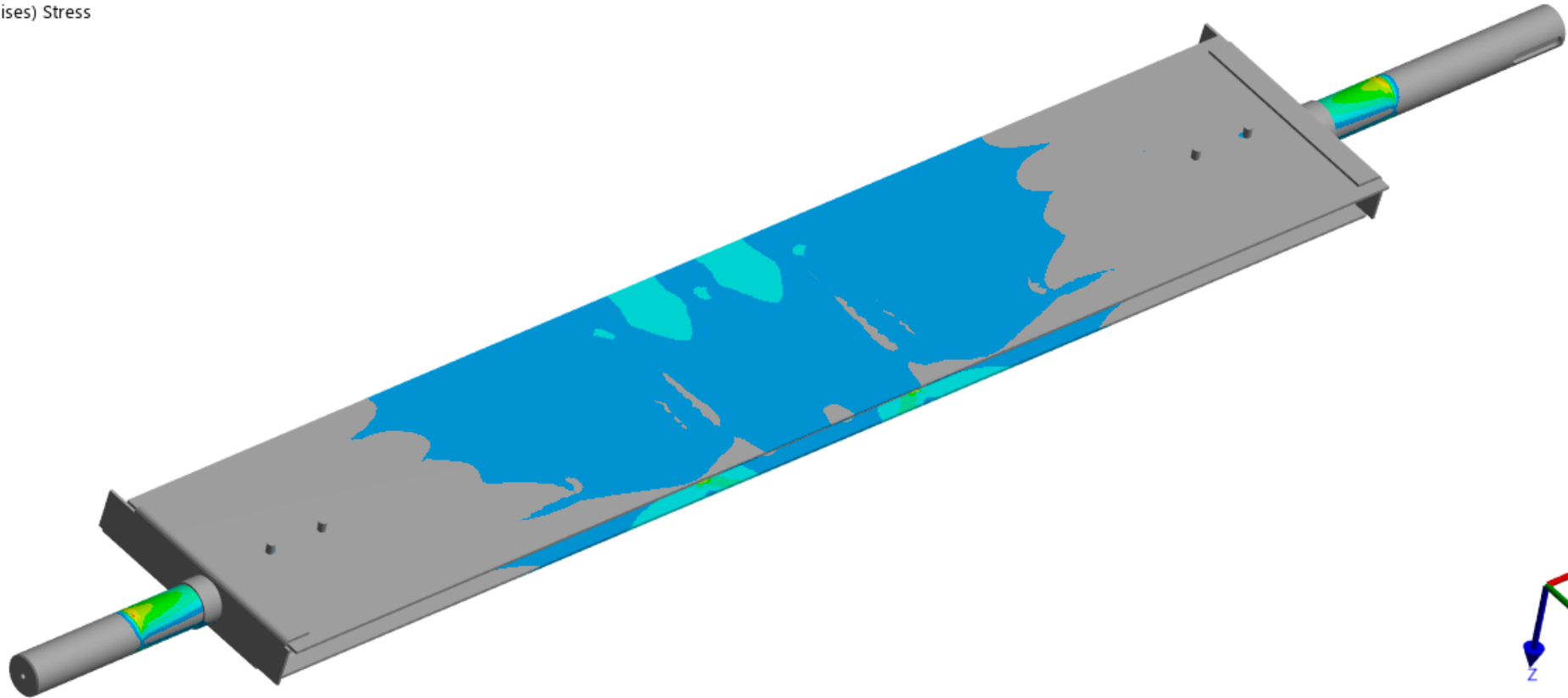
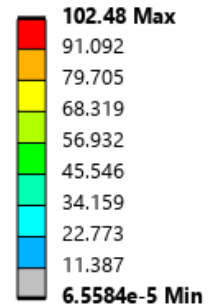
B: 6148 blade self weight+pressure

Equivalent Stress

Type: Equivalent (von-Mises) Stress

Unit: MPa

Time: 1 s



Max. Von-Mises Stress = 102 Mpa

Stresses in the Blade are within the materials allowable stress limit (i.e. 140 Mpa).



From the above FEA analysis it is been observed that,

- Total deformation in the Blade is **1.7 mm**.
- Von-mises Stresses in the Blade assembly are well **within** the materials allowable Stress limits (i.e. 140 Mpa).
- Thus, the blade is structurally **adequate to withstand** the applied load under this load case.



Damper Blade

Case-2 : Self Weight + Pressure + Moment

4. Boundary Conditions :

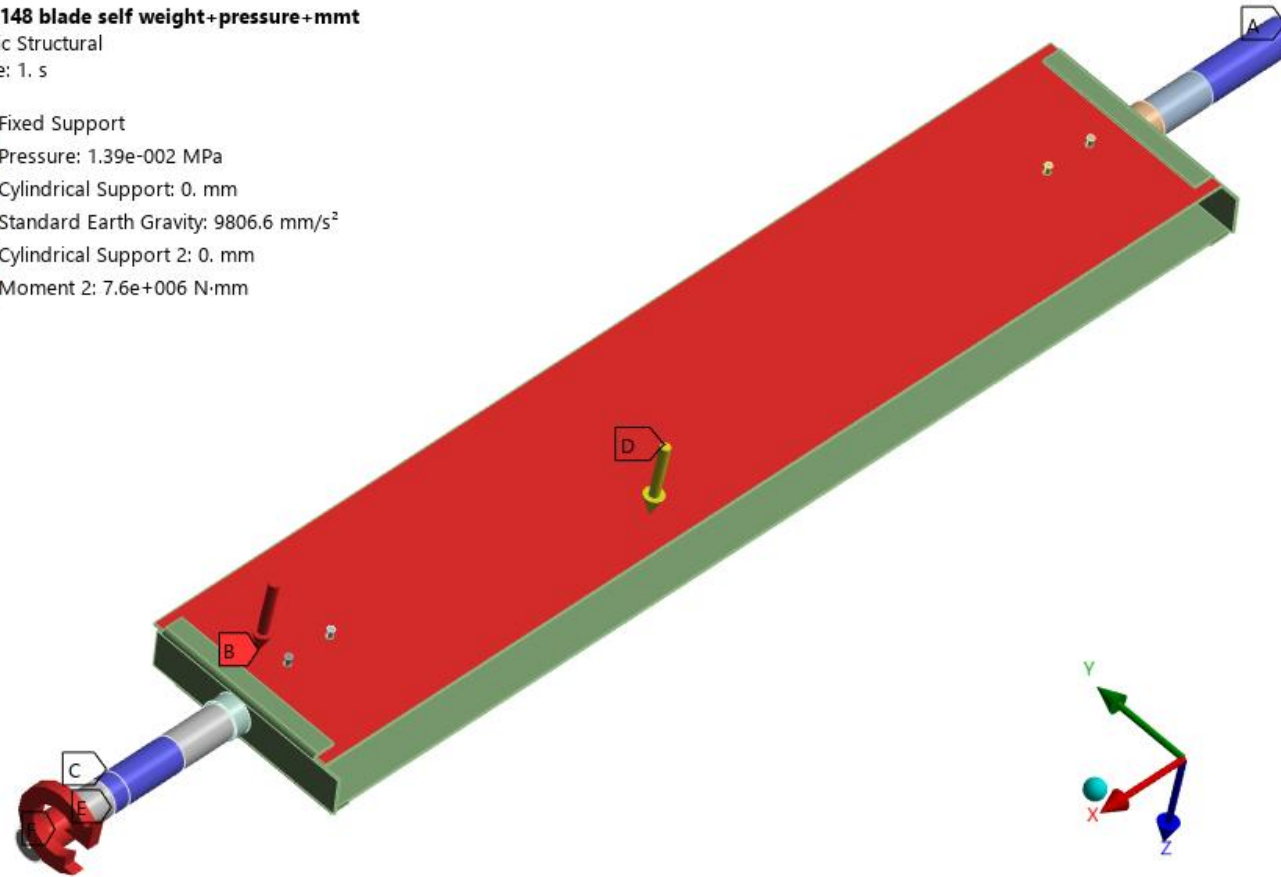


A: 6148 blade self weight+pressure+mmt

Static Structural

Time: 1. s

- A** Fixed Support
- B** Pressure: 1.39e-002 MPa
- C** Cylindrical Support: 0. mm
- D** Standard Earth Gravity: 9806.6 mm/s²
- E** Cylindrical Support 2: 0. mm
- F** Moment 2: 7.6e+006 N-mm



A) Fixed Support.

B) Design Pressure of 1421.92 mmWc which is 0.0139Mpa.

C,E) Driving & Non-driving shaft is free in Rotation.

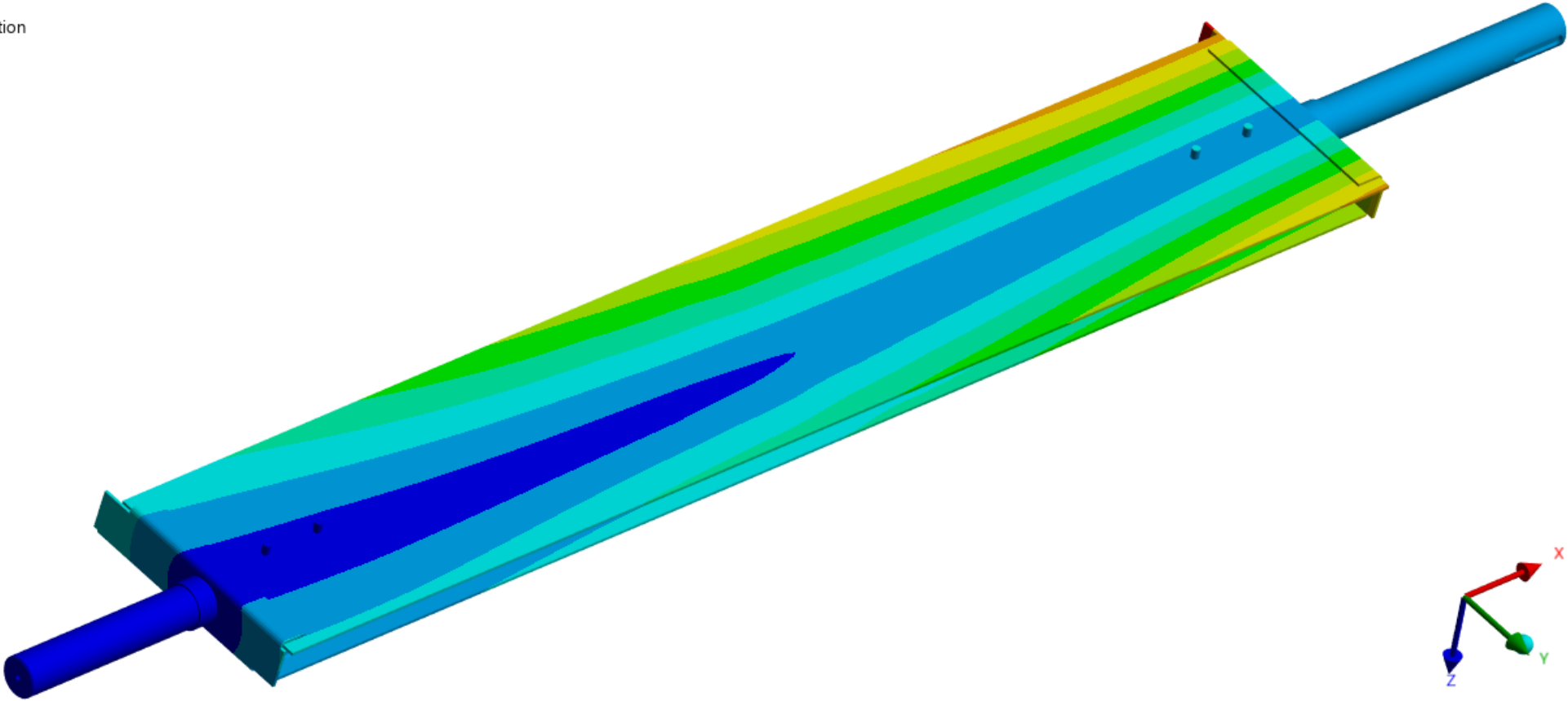
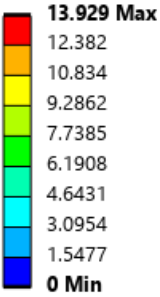
D) Self weight is been considered.

F) Moment of 7.6E+06 N*mm (per blade) is been applied to driving shaft.

5. Results : Deformation Plot



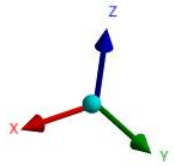
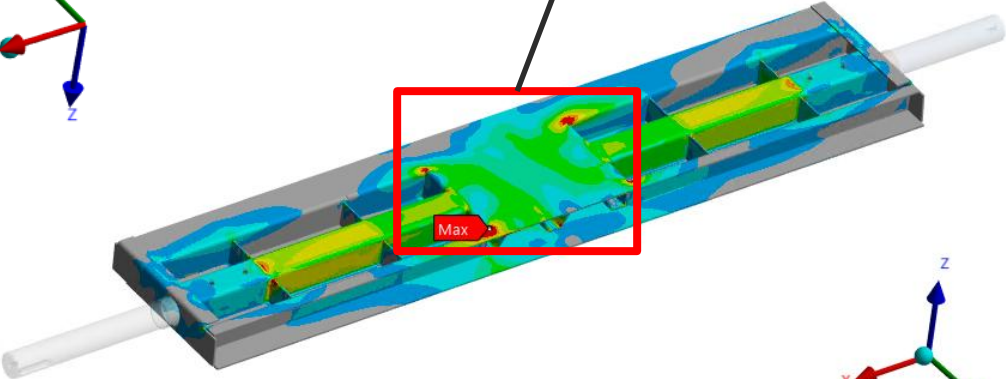
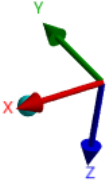
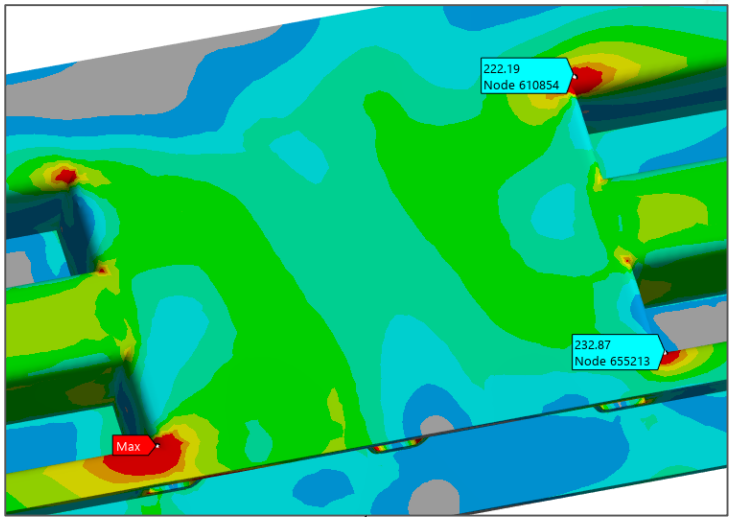
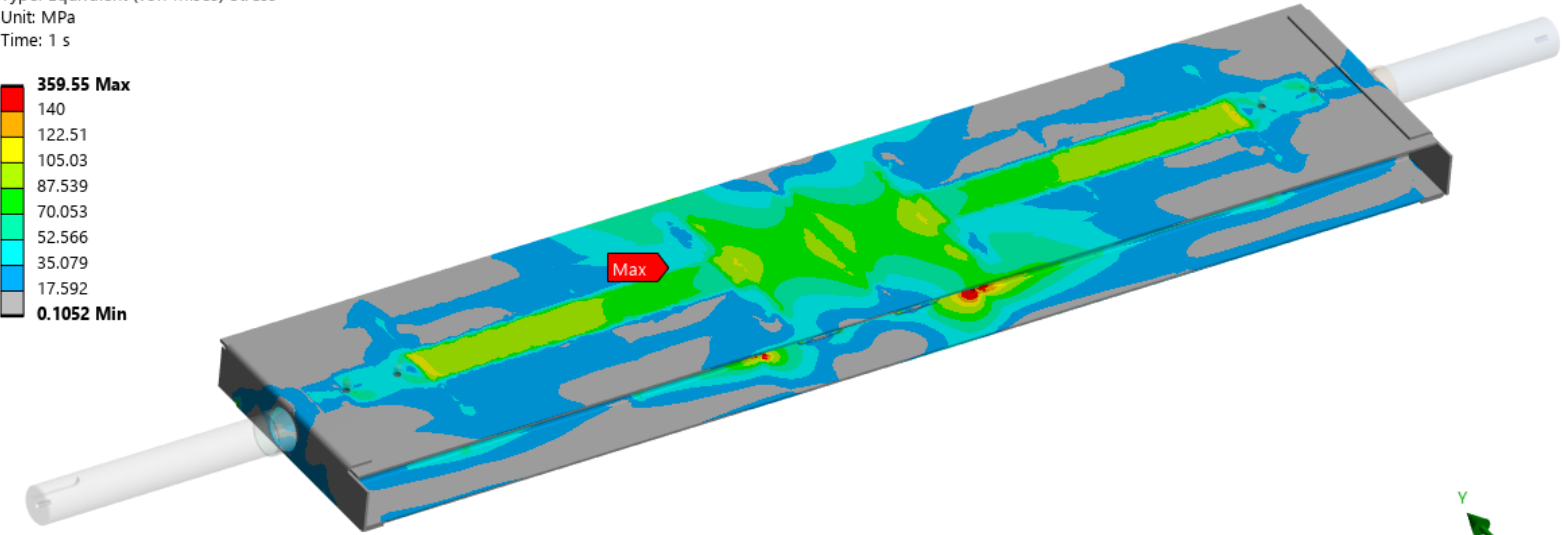
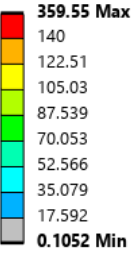
A: 6148 blade self weight+pressure+mmt
Total Deformation
Type: Total Deformation
Unit: mm
Time: 1 s



Max. Total Deformation = 14mm

5.1 Results : Von-mises Stress Plot

A: 6148 blade self weight+pressure+mmt
Equivalent Stress
Type: Equivalent (von-Mises) Stress
Unit: MPa
Time: 1 s



Max. Von-Mises Stress = 359 Mpa

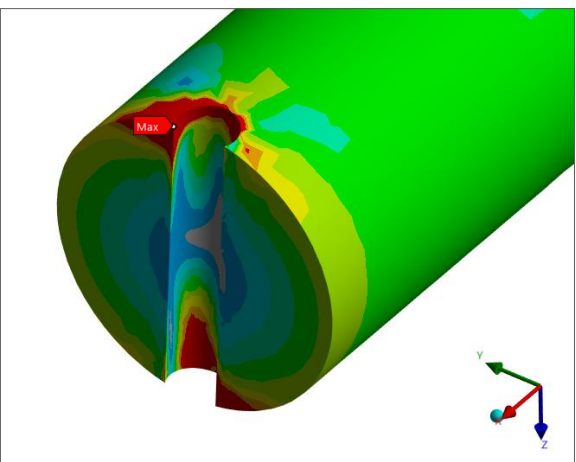
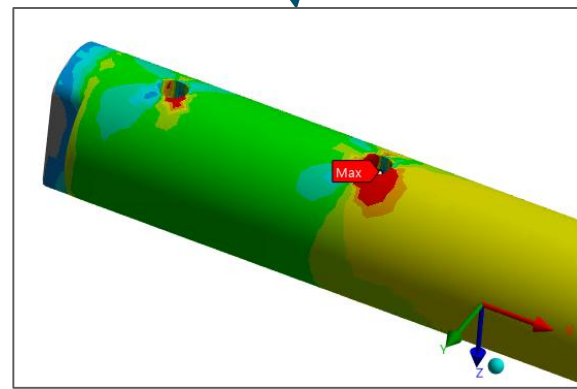
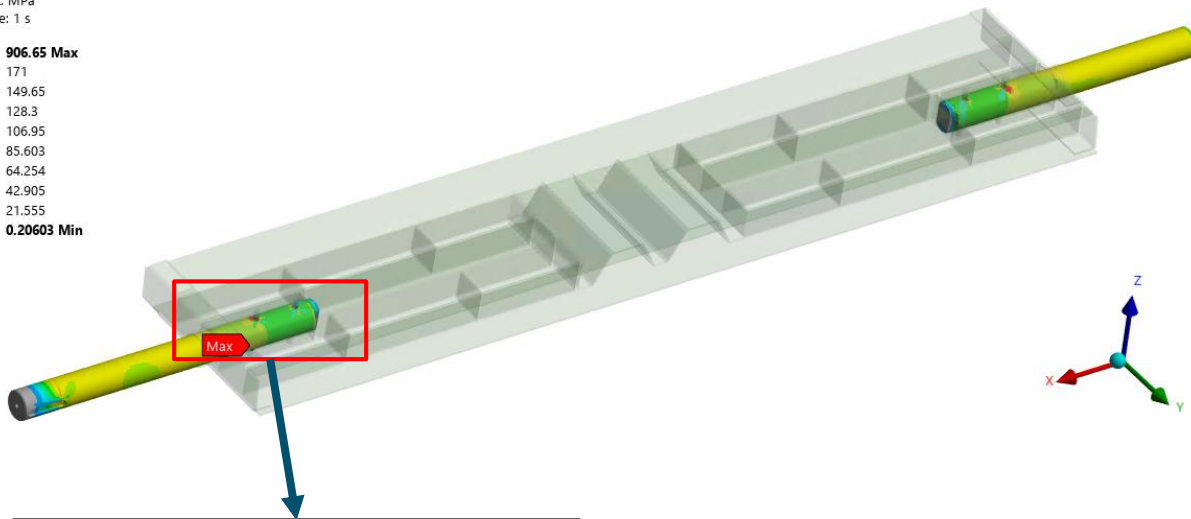
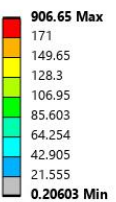
Stresses in the Blade are exceeding the materials allowable Stress limit (i.e. 140 Mpa).

5.2 Results : Shaft



Von-mises stress

A: 6148 blade self weight+pressure+mmt
Equivalent Stress Shaft
Type: Equivalent (von-Mises) Stress
Unit: MPa
Time: 1 s



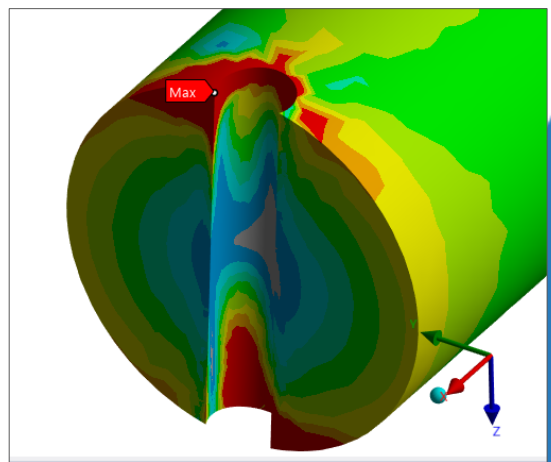
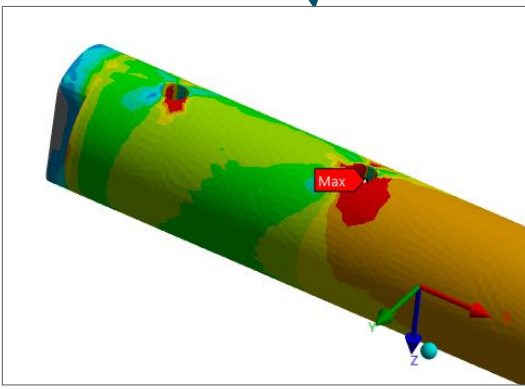
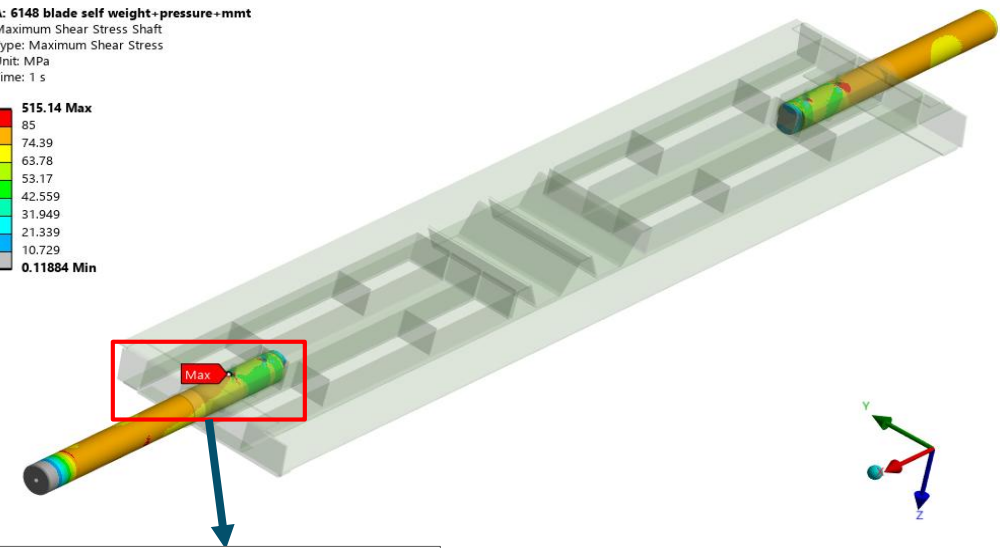
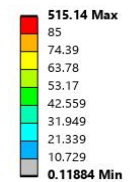
Max. Von-mises Stress = 906Mpa

Cut section

Max. Von mises Stresses in the Shaft are exceeding the allowable stress limit (i.e.171 Mpa).

Max-shear stress

A: 6148 blade self weight+pressure+mmt
Maximum Shear Stress Shaft
Type: Maximum Shear Stress
Unit: MPa
Time: 1 s

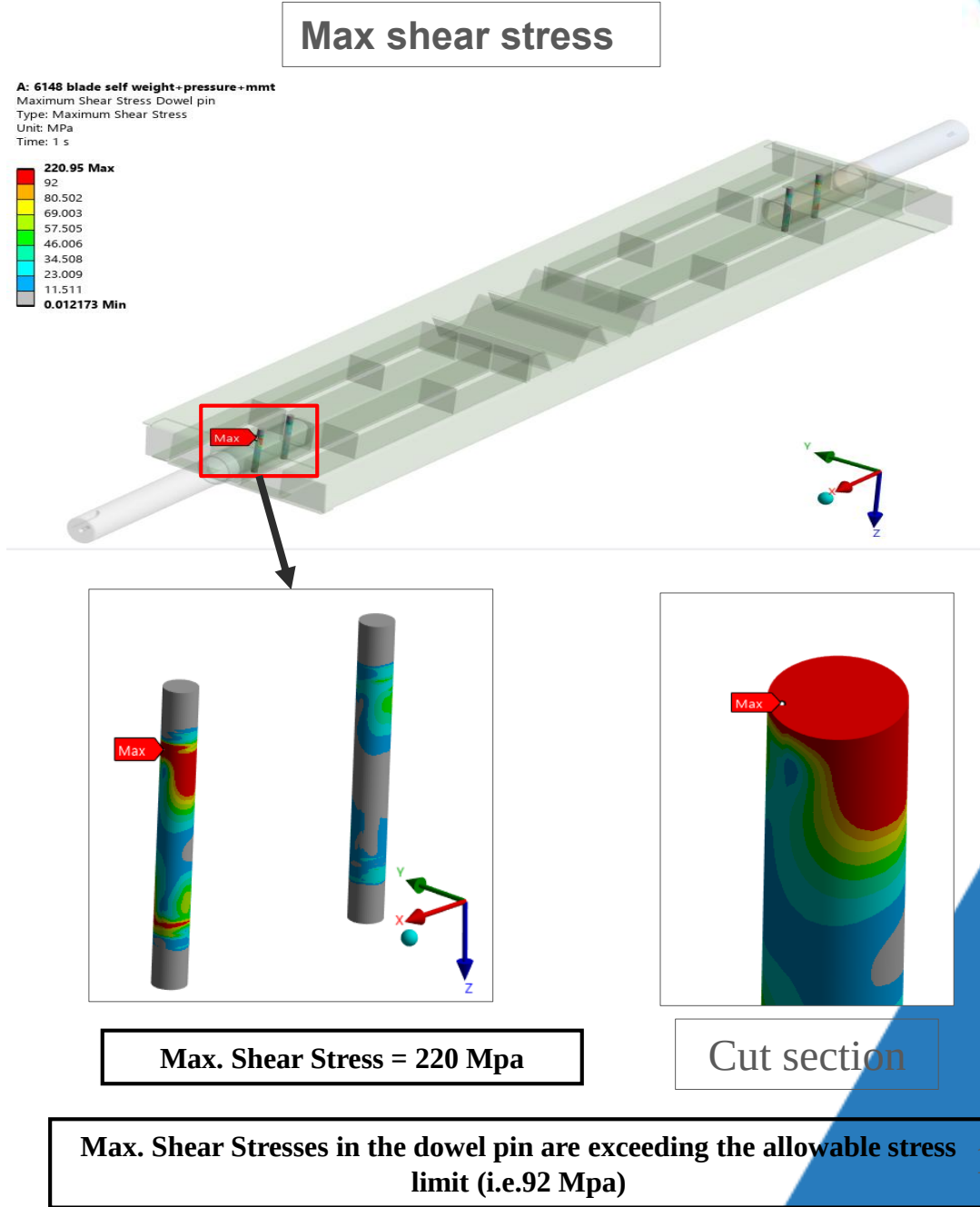
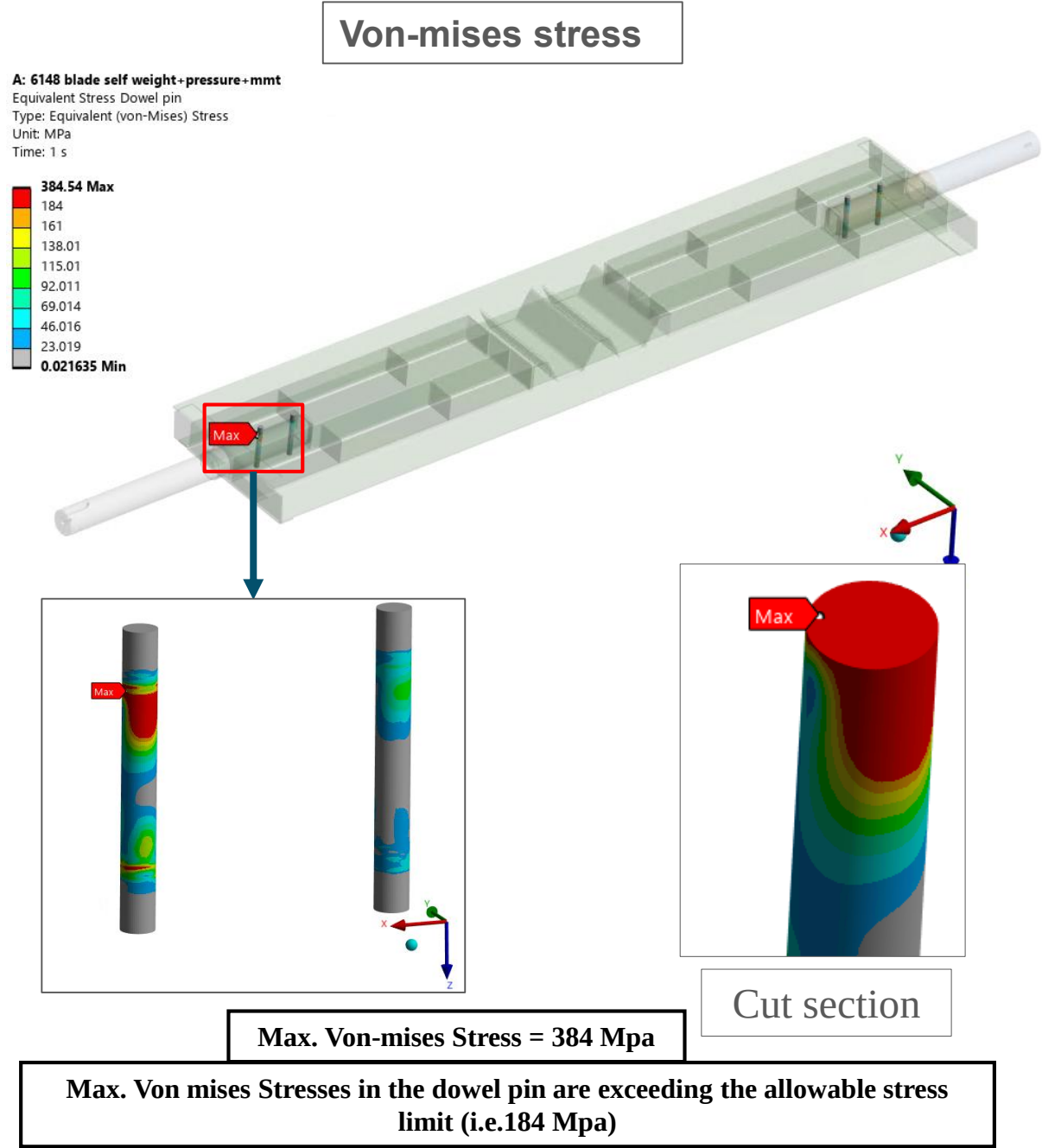


Max. Shear Stress = 515 Mpa

Cut section

Max. Shear Stresses in the Shaft are exceeding the allowable stress limit (i.e.85 Mpa).

5.3 Results : Dowel pin



6. Observation & Summary :



From the above FEA analysis it is been observed that,

- Total deformation in the Blade is **14mm**.
- Von-mises Stresses in the **Blade** are **exceeding** the materials allowable Stress limit (i.e. 140 Mpa).
- Max. Von-mises & Shear Stresses in the **Shaft are exceeding** the allowable Stress limit.(i.e. 171 Mpa & 85 Mpa).
- Max. Von-mises & Shear Stresses in the **Dowel pins are exceeding** the allowable Stress limit.(i.e. 184 Mpa & 92 Mpa).
- Thus, the blade assembly is **not structurally adequate to withstand** the applied load under this load case.



THANK YOU

